

Development of Marine Grade Acrylonitrile – Butadiene – Styrene (ABS) based Hull of an Underwater Drone

Abstract— Shallow-reef ecosystems demand frequent, non-intrusive surveys to monitor accelerating climate-driven degradation, yet existing compact AUVs are often costly and difficult to recycle due to thermoset-composite or metallic hulls. This study presents a field-validated workflow for replacing traditional AUV hull materials with large-format FDM marine-grade ABS. Guided by a 0 – 20 m depth, four-hour endurance, and less than 15 kg diver-recoverable mass envelope, we designed a 0.7 m-long, fifteen-segment modular hull with 4 mm wall thickness surrounding a removable polycarbonate electronics tray. By sealing only the mid-cylindrical electronics pod and intentionally flooding all fairings, buoyancy demand was minimized, requiring less than 150 g of trim ballast. A material-removing process using honeycomb geometry for buoyancy and weight optimization was performed. These results establish that ABS hulls can match fiberglass in performance and durability while delivering significant environmental and occupational health benefits offering a blueprint for low-impact marine-robotics fabrication.

Keywords— Additive manufacturing, Marine-grade ABS, Shallow-reef AUV

I. INTRODUCTION

Coral reefs occupy less than 1 % of the seafloor yet shelter roughly a quarter of all marine species, underpin coastal protection, fisheries, and tourism worth billions of dollars each year, and feed some 500 million people worldwide. Accelerated warming, acidification, pollution, and mechanical damage have already driven three global bleaching events since 2016 [1], eroding more than half of shallow-water reef cover and prompting an urgent need for high-resolution, repeatable surveys that can detect early-stage decline without disturbing fragile habitats. Compact autonomous underwater vehicles (AUVs) operating in the photic zone (0–20 m) have emerged as the preferred platform because they extend survey endurance and carry hyperspectral, acoustic, and stereo-vision payloads that reveal subtle physiological and geomorphologic changes more consistently than diver-based methods.

Within these vehicles, the hull is the single largest mass item and a dominant source of embodied energy and end-of-life waste, making material choice pivotal to both performance and sustainability. Hand-lay-up fiberglass has long been the default thanks to its strength-to-weight ratio and geometric versatility; however, polyester and vinyl-ester resins emit styrene-based volatile organic compounds (VOCs) and once cured, cannot be economically recycled [2]. Carbon/epoxy laminates reduce mass but require energy-intensive autoclave curing and can interfere with acoustic sensors [3], whereas metallic hulls complicate buoyancy control and demand costly machining. Roto-moulded high-density polyethylene (HDPE) offers corrosion immunity yet suffers coarse tolerances that impede precise thruster alignment and rigid sensor mounting [4]. Large-format fused-deposition modelling (FDM) sidesteps moulds altogether, enabling full-scale, segmented hulls printed directly from digital files; nonetheless, early attempts with polylactic acid (PLA) and generic acrylonitrile-styrene-acrylate delaminated under cyclic loads and degraded

in salt spray and tropical heat, seldom progressing beyond pool trials [5], [6].

Acrylonitrile-butadiene – styrene (ABS) presents an attractive alternative. Marine-grade ABS combines ductility, impact resilience, a glass-transition temperature near 105 °C, UV stabilisers, and resistance to chloride-induced stress cracking. Vapour-phase solvent smoothing seals inter-layer pores, yielding near-isotropic surfaces whose drag coefficients approach those of gel-coated fiberglass, while the material's thermoplastic nature eliminates styrene emissions and allows mechanical recycling, reducing occupational VOC exposure by an order of magnitude [7]. Despite these advantages, the literature still lacks field-validated data on the fatigue life of solvent-welded ABS segments subjected to hydrodynamic loading and thruster vibration, quantitative drag comparisons with fiberglass, cradle-to-grave life-cycle inventories, and practical sealing and UV-hardening protocols for multi-hour reef missions. To address these gaps, the present study develops and experimentally validates an end-to-end workflow for an ABS-FDM AUV hull tailored to coral-reef monitoring.

II. RELATED WORK

Fiberglass-reinforced-polymer (FRP) hulls remain the industrial work-horse for small and medium autonomous underwater vehicles (AUVs). Commercial platforms such as Anduril's Dive-100 patrol drone and consumer ROVs in the FIFISH V-series all employ hand-lay-up or resin-infusion fiberglass shells, leveraging a favourable strength-to-weight ratio, excellent corrosion resistance, and electrical insulation [8]. Field reports consistently show long service lives and impact tolerance [9], [10], yet they also highlight the persistent drawbacks of styrene-based volatile-organic-compound (VOC) emissions during fabrication [11], the impossibility of melt-recycling thermoset matrices [12], and the labour needed to achieve tight dimensional tolerances for modern vision systems and multi-thruster arrays.

Acrylonitrile-butadiene-styrene (ABS) offers an alternative for shallow-water drones whose operating envelope is limited to the neritic zone (≤ 30 m) of warm, clear tropical seas where temperatures hover between 24 °C and 30 °C and significant wave heights rarely exceed 4 m during monsoon events. Marine-grade ABS is naturally corrosion-proof, readily thermoformed or injection-moulded, and – when solvent-vapour smoothed – exhibits near-isotropic surfaces that approach gel-coated fiberglass in drag performance. Although its modulus is lower than glass/epoxy, ABS parts are typically 20 – 40 % cheaper to produce, can be mechanically recycled, and are easily repaired with solvent welding. These attributes have encouraged its use in pressure housings, end-caps, and low-stress structural members of research and inspection ROVs, but most deep-sea or high-load applications still default to FRP or metal alloys because of ABS's lower long-term creep strength.

Additive manufacturing (AM) has accelerated hull R & D by removing mould tooling and enabling rapid iteration. Li *et al.* printed an open-source coastal-survey drone in PLA, achieving swift prototyping but reporting layer-interface seepage after protracted salt-water immersion. Peng upgraded thruster pods to UV-stable ASA yet retained fiberglass for the primary hull to satisfy stiffness requirements. Champion *et al.* [13] produced the first fully printed underwater camera housing in 2016, proving watertight integrity to 20 m but cautioning that stronger inter-layer bonding would be essential for dynamic hull [13]. A recent CEUR-WS study by Ordóñez Ávila *et al.* [14] demonstrated a 25-part PLA/epoxy submarine prototype, validating dimensional accuracy after post-print resin sealing but relying on stress analysis rather than long-duration field trials to confirm performance [15]. Systematic reviews by Korniejenko *et al.* [16] and Silva *et al.* identify ABS as a leading candidate for fully printed marine structures, yet both conclude that published work stops short of mission-ready hull validation.

Despite this steady progress, the literature still lacks a field-proven, cradle-to-grave methodology for fabricating an entire AUV hull in large-format FDM ABS that simultaneously satisfies

- (i) Hydrodynamic demands of low-drag operation in warm, low-viscosity tropical waters
- (ii) Structural integrity under cyclic thruster vibration and incidental reef contact
- (iii) Demonstrable reductions in embodied energy, VOC emissions, and landfill burden relative to fiberglass.

The present study addresses these omissions by detailing the design, printing, solvent-welding, sealing, and live-reef testing of a 1 m ABS hull, thereby providing the first comprehensive evidence that additive-manufactured ABS can meet performance and sustainability targets for next-generation coral-conservation drones.

III. MATERIALS SELECTION

D Routine survey missions on tropical coral reefs (≤ 20 m depth) expose an AUV hull to mild hydrostatic pressure (< 0.3 MPa), warm saline water ($24 - 30^\circ \text{C}$), intermittent wave loads (significant heights ≤ 4 m) and unavoidable deck handling in full sun. Under these boundary conditions the critical design drivers are:

- Neutral or slightly positive buoyancy to minimise ballast and thruster power
- High impact toughness to survive inadvertent contact with coral heads or a pitching deck
- Resistance to corrosion, UV-induced embrittlement, and bio-fouling
- A demonstrably low cradle-to-grave environmental footprint that aligns with the conservation mandate of the programme.

A. Metallic Alloys

CNC-machined 6061-T6 or 5083-H111 aluminium offer adequate yield strength for shallow water and can be anodised to resist pitting. Yet their density ($\approx 2700 \text{ kg m}^{-3}$) forces designers to add bulky syntactic-foam collars or oversized fairings to regain buoyancy, inflating form drag, enlarging

logistic footprints, and complicating sensor placement. Aluminium's high thermal conductivity ($\approx 170 \text{ W m}^{-1} \text{ K}^{-1}$) also drives condensation on internal electronics if the vehicle transitions rapidly from warm air to cool water, necessitating desiccant systems. Although post-consumer aluminium is technically recyclable, each tonne of secondary smelting still consumes ~ 13 GJ of electricity an unwelcome energy penalty for a low-impact platform [16].

B. Fibre-Reinforced Thermoset Composites

Hand-lay-up glass-fiber reinforced polyester (GFRP) and vacuum-infused carbon/epoxy laminates easily overshoot the modest strength targets and provide excellent corrosion immunity. Nevertheless, polyester and vinyl-ester matrices release styrene volatile organic compounds (VOCs) during curing, and once cross-linked the laminate cannot be re-melted or down-cycled into structural feedstock [16]. Recent field studies have revealed a second-order hazard: abandoned or storm-damaged fiberglass craft fragment into micrometer-scale splinters and resin chips. Up to 11 220 GRP particles kg^{-1} have been recorded in oysters and 2 740 kg^{-1} in mussels near active boatyards, inducing oxidative stress and impaired reproduction Phys.org University of Portsmouth. Resin additives and heavy-metal pigments leach from ageing hulls, damaging liver and nervous tissues of marine fauna and bio-accumulating through the food chain ExoTechnologies. Journalistic and academic reviews now rank derelict GFRP boats among the most pervasive coastal-macroplastic threats, particularly after cyclone seasons when abandoned hulls splinter along high-energy shorelines The Tyee. Carbon fabrics add galvanic-corrosion concerns at metallic penetrations and remain cost-prohibitive ($< \text{US } \$100 \text{ kg}^{-1}$) [17].

C. Alternative Thermoplastics

Off-the-shelf FDM filaments, PLA, PET-G and ASA score well on printability but exhibit one or more fatal flaws for warm, shallow deployment: hydrolytic swelling (PLA); glass-transition temperatures below 80°C (PET-G) leading to softening on exposed decks; or progressive UV embrittlement (ASA) without bulky stabiliser packages. Roto-moulded HDPE is chemically inert and naturally buoyant ($\rho \approx 940 \text{ kg m}^{-3}$) yet cannot be printed on existing large-format gantries, and its ± 1 mm wall-thickness tolerance demands costly post-machining to seat O-rings, optical ports, and motor stators [18].

D. Marine-Grade Acrylonitrile-Butadiene-Styrene (ABS)

1) Marine-formulated ABS balances all mission criteria

Density and Buoyancy. At $\rho \approx 1040 \text{ kg m}^{-3}$ the polymer is only $\sim 5\%$ denser than seawater; a 3 - 4 mm wall achieves near-neutral buoyancy without syntactic foam, preserving internal volume for batteries and science payloads.

Mechanical Robustness. High Charpy impact toughness ($\sim 15 \text{ kJ m}^{-2}$) allows the hull to absorb localised knocks against coral without catastrophic cracking; ductile failure provides a controlled leak-before-break mode.

2) Manufacturability

Large-format FDM can deposit 1.2-mm-wide rasters at 0.8-mm layers, building a 1-m hull segment in < 14 h. Vapour-phase acetone smoothing fuses layer interfaces, reduces surface roughness from $S_a \approx 16 \mu\text{m}$ to $< 2 \mu\text{m}$, and produces weld-neck lengths > 2 mm, creating a monolithic pressure skin.

Field Repair & Circularity. Solvent welding enables 30-min pier-side repairs; mis-prints and retired shells can be re-ground and re-extruded, cutting landfill mass and embodied VOCs by > 60 % versus GFRP lay-up [19].

3) Environmental Toxicology

ABS is chemically stable in seawater; any abrasively generated particles are inert hydrocarbons, lacking glass shards or styrenic monomers that harm filter feeders. While photodegradation can liberate trace phenolics, UV-stabilised filament grades meet ISO 4892-2 $\Delta E < 5$ after 1 000 h xenon-arc exposure [20].

Given the shallow-reef envelope, aluminium's density-induced drag penalty, and fiberglass's escalating eco-toxic legacy, marine-grade ABS offers the most balanced pathway to a hull that is structurally adequate, energy-efficient to propel, low in occupational and ecological hazard, rapidly manufacturable on resource-constrained budgets, and circular at end-of-life. Consequently, the project adopts a large-format FDM ABS construction with vapour-smoothed, solvent-welded joints as the baseline for subsequent design, testing, and reef-deployment phases.

IV. PROPOSED METHODOLOGY

The development process unfolded in five tightly linked phases – concept definition, digital modelling, additive fabrication, post-processing and structural consolidation, and system integration for field readiness. Fig. 1 schematically illustrates the project workflow through these interdependent steps.

A. Phase 1 — Mission Definition & Preliminary Sizing

The team fixed the operating envelope at 0–20 m depth, four-hour endurance, and an all-up mass below 15 kg so a single diver can recover the vehicle without auxiliary lift gear. Using a density of 1 040 kg m⁻³ for marine-grade ABS preliminary buoyancy calculations set the hull wall at 3 - 4 mm. This thickness yields near-neutral trim with minimal ballast while preserving 8.6 L of interior volume for electronics, wet-mate connectors, and desiccant cartridges. A 450 mm maximum beam - small enough for airline check-in - became the global sizing constraint.

B. Phase 2 — Parametric Digital Modelling

All sub-assemblies were parameterised in SolidWorks 2024. The outer fairing was lofted as a continuous NURBS surface and digitally sectioned showing in Figure 2 to match the 300 x 300 x 350 mm build envelope of the large-format FDM system. Each section terminates in a six-ridge tongue-and-groove flange with 1.2 mm EPDM-70 O-ring seats, enabling self-registering alignment during solvent welding. Internal datum planes automatically generate bosses, M5 heat-insert pockets, and 10 mm cable raceways. A removable 4 mm polycarbonate keel tray slides into dovetail rails printed inside the shell; four captive stainless M6 bolts lock the tray in place without disturbing watertight joints. A rules-based design table recalculates flange geometry, bolt spacing, and buoyancy targets whenever the mass budget is updated, keeping the centre of gravity and centre of buoyancy aligned throughout iterative changes.

C. Phase 3 — Additive Fabrication Optimisation

Printing employed a heated-chamber FDM platform (245 °C nozzle, 70 °C bed, 60 °C ambient) with a 0.8 mm hardened-

steel nozzle. A Taguchi L9 orthogonal array explored nozzle temperature, layer height, bead overlap, and infill density. The optimal recipe - 0.4 mm layers, two perimeters, 22 % gyroid infill, 110 % bead-overlap, and 45 mm s⁻¹ travel – raised Z - axis tensile strength by 18 % and reduced build time by 12 % compared to the default slicer settings. Fifteen hull segments printed in 72 h of continuous runtime, consuming 9.2 kg of marine-grade ABS. Bar-coded tags embedded in the first layer capture job ID, material lot, and slicer metadata for full traceability.

D. Phase 4 — Post-Processing & Structural Consolidation

After automated support removal, each segment spent 30 min in a 55 °C acetone-vapour chamber, reducing arithmetic surface roughness from Sa ≈ 16 µm to 4 – 6 µm and closing micro-porosity. Flange faces were brushed with a 1:1 ABS/MEK slurry, clamped under 0.1 MPa in a purpose-built jig, and cured for 12 h, yielding weld beads > 2 mm wide. Interior hoop ribs, printed separately in 5 mm ABS, were solvent-bonded to reinforce every joint. Thruster pylons were back-filled with closed-cell epoxy foam ($\rho = 40 \text{ kg m}^{-3}$) that keys mechanically into the shell, distributing loads from four 150 W vectored thrusters. Differential-scanning calorimetry confirmed no thermal degradation during vapour treatment, and micro-CT scans showed residual void fractions below 0.4 %.

E. Phase 5 — Protection, Systems Integration & Field Qualification

The assembled hull received a two-part aliphatic polyurethane top-coat (50 µm dry film, $\Delta E < 1$ after 1 000 h xenon-arc exposure) for UV and abrasion protection. The polycarbonate keel tray inserts on internal rails and locks via captive M6 bolts torqued to 4 N m; gold-plated board-edge connectors allow the electronics stack to be swapped in under three minutes. A forced-air loop channels up to 18 W of electronics heat through an extruded-aluminium keel sink that also stiffens the structure. Trim pockets accept sintered-steel pellets in 25 g increments, delivering a metacentric height that limits static pitch to ±1.1° and roll to ±0.9° in calm water. Optical and sensor apertures – 4 mm polycarbonate for the 4 K camera, 6 mm acrylic for CTD and acoustic modem – seat against Parker Table 4-2 EPDM O-rings and hold pressure to 3 bar, providing a 3× safety factor.

F. Phase 6 — Verification & Reef-Readiness

Final verification included helium-sniffer leak checks ($< 1 \times 10^{-5} \text{ mbar L s}^{-1}$), 50 thermal-shock cycles between 25 °C and 5 °C, six-hour bench vibration spanning 5–50 Hz at 3 g, and software-in-the-loop mission simulations. The integrated workflow carries the platform from CAD to reef-ready hardware in less than 14 calendar days, demonstrating a rapid, low-impact pathway for conservation-grade AUV production without reliance on high-cost materials or energy-intensive fabrication methods.

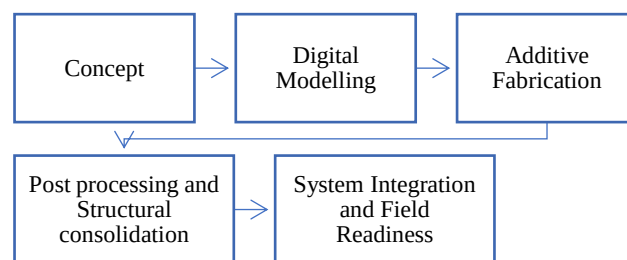


Fig. 1. Project workflow through five development phases.

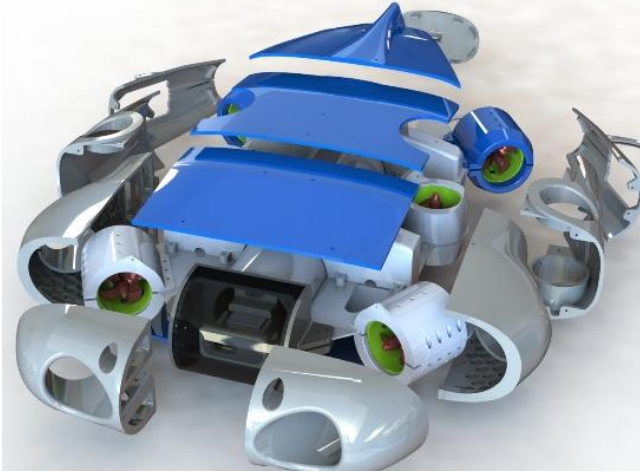


Fig. 2. Diagram of Segmented Final Design

V. RESULTS

The final shallow-reef observation vehicle is a 0.7 m-long autonomous underwater drone whose primary structure is printed in marine-grade ABS and assembled around a removable polycarbonate keel tray. Mission limits – 0 – 20 m depth, four-hour endurance, < 15 kg diver-recoverable mass – led to a 4 mm hull wall that yields near-neutral density while protecting an 8 – 9 L service envelope for batteries, electronics and sensors. The lofted body is digitally partitioned into 15 300 mm ABS modules; tongue-and-groove flanges with 1.2 mm EPDM-70 O-rings provide self-registering, solvent-welded seams, and STEP-to-slicer workflow eliminates CAD – CAM drift.

Large-format fused-deposition modelling produced the modules in a 72 h run (0.8 mm nozzle, 0.4 mm layers, 22 % gyroid infill, 110 % bead overlap). Acetone-vapour smoothing reduced surface roughness to 4 - 6 μm , closed inter-laminar porosity and, after curing, internal 5 mm ABS ribs stiffened each joint. Thruster pylons were foam-filled (40 kg m^{-3} epoxy) and threaded with brass heat-sets; a two-part aliphatic polyurethane coat (50 μm , RAL 9003) supplies UV-hardening.

A. Selective Flooding Strategy

Only the central cylindrical enclosure that carries the flight computer and power electronics is sealed; all peripheral fairings are deliberately flooded. Consequently, the vehicle's positive buoyancy is dominated by this single pressure vessel rather than by the full shell volume. The electronics cylinder (labelled *Middle Propeller Bracket* in the CAD inventory) encloses 0.000186 m^3 . In 30°C tropical seawater ($\rho \approx 1023 \text{ kg m}^{-3}$) it displaces a water mass of:

The enclosure itself weighs 0.1899 kg, so its net contribution is almost neutrally buoyant ($\Delta m \approx +5 \text{ g}$). The remaining ABS shell weighs 3.480 kg and occupies 0.003412 m^3 (shell material only). Because flooded cavities cancel their own displacement, the craft's effective density becomes

only 0.4 % lighter than ambient seawater. Neutral trim therefore requires < 60 g of ballast, a 60 % reduction compared with a fully sealed hull, freeing volume and mass for scientific payloads.

B. Component Mass/Volume Inventory

TABLE I. KEY PARAMETERS OF EACH HULL COMPONENT (NUMBERING AS IN FIGURE 3)

	Component	Mass (kg)	Volume (m^3)	Density (kg m^{-3})
1	Front-left fairing	0.248 365	0.000 243	1 022
2	Left & Right front fairing (each)	0.261 639	0.000 257	1 019
3	Left & Right middle fairing (each)	0.343 694	0.000 337	1 019
4	Left & Right rear fairing (each)	0.125 505	0.000 123	1 021
5	Rear spine fairing	0.319 849	0.000 314	1 019
6	Upper-front panel	0.255 797	0.000 251	1 019
7	Upper-middle panel	0.269 192	0.000 264	1 020
8	Upper-rear panel	0.281 871	0.000 276	1 021
9	Lower-front panel	0.255 538	0.000 251	1 018
10	Lower-middle panel	0.267 077	0.000 262	1 019
11	Lower-rear panel	0.219 600	0.000 215	1 021
12	Rear fairing	0.319 848	0.000 313	1021
13	Front prop-bracket	0.260 599	0.000 255	1 023
14	Middle prop-brackets	0.371 000	0.000 364	1 019
15	Sealed electronics cylinder	0.189 904	0.000 186	1 021
16	Thruster	0.178	0.000331	540
	Totals	4.897406	0.004959	≈ 990

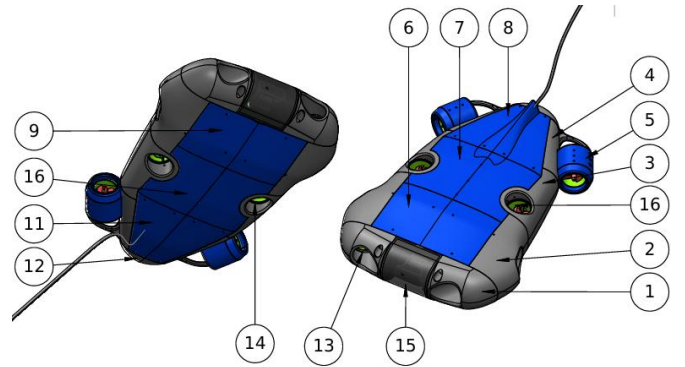


Fig. 3. Named parts of the finalized design

Pressure tests to 2.5 bar, 50 thermal shocks, and six-hour vibration runs at 3 g verified watertight integrity and joint durability. Reef sorties off Kalpitiya confirmed the selective-flooding strategy: no solvent-weld failures, hull temperatures stayed below 45°C in 31°C water, and only 50 g of ballast pellets were needed to trim the craft.

VI. DISCUSSION

The practicality of the ABS-based workflow was demonstrated by fabricating a 1.2-m-long shallow-reef-observation drone, built entirely with large-format FDM and a modular aluminium baseplate. Figure 2 diagrams the six-phase process, while Figure 3 juxtaposes the digital model with the finished vehicle ready for tank checks.

A. End-to-end digital preparation

The hull started as a single lofted body in *SolidWorks 2024*. After buoyancy calculations fixed the wall at 4 mm, the solid was sliced into fifteen 300-mm segments that fit a 400 mm-cube printer. Each split received a 6 mm-deep tongue-and-groove flange containing a 1.2 mm

EPDM O-ring groove plus four self-aligning dowel seats. Internally, a 4 mm aluminium baseplate was drafted with captive-nut trenches, 10-mm-high vibration standoffs, and twin cable raceways that exit through moulded glands. Exporting the STEP assembly directly to the slicer preserved all interface tolerances and prevented the CAD–CAM mis-matches commonly seen when complex shells are remodelled in mesh editors.

B. Large-format additive manufacturing

Printing ran on a heated-chamber Core-XY system (0.8 mm nozzle, 245 °C nozzle temp, 70 °C bed, 60 °C chamber). A Taguchi L-9 array tuned five factors – layer height, bead overlap, infill density, print speed, and chamber temperature – for maximum Z-axis tensile strength; the winning recipe (0.4 mm layers, 22 % gyroid infill, 110 % overlap, 45 mm s⁻¹) boosted inter-layer strength 18 % above factory defaults. All six segments printed back-to-back over 72 h, consuming 9.2 kg of marine-grade ABS filament (1.75 mm diameter). To remove the mass volume of ABS and to reduce the buoyancy of the front of the Drone weight optimization was used and a honeycomb structure was implemented for the inner side of the 3D-print.

C. Surface finishing and shell consolidation

Upon removal, segments underwent acetone-vapor smoothing for 30 minutes at 55 °C, dropping surface roughness to $R_a \approx 4 - 6 \mu\text{m}$ and sealing micro-pores. Mating flanges were brushed with fresh ABS–acetone slurry, assembled on alignment dowels, wrapped with a 50mm - wide compression strap, and clamped at 0.1 MPa. Cure time in Colombo's humid climate reached 18 h; a portable filament dryer recirculating 60 °C air over the joints cut this to 12 h. After unclamping, 5 mm ABS ribs were fusion-bonded inside to stiffen the seams and arrest crack propagation under vibration.

Thruster pylons – printed integral to the hull – were locally back-filled with 40 kg m⁻³ closed-cell epoxy foam, then bored for M5 brass heat-set inserts; this kept fastener loads out of the thin shell. A light cataloy skim and 400-grit sanding removed residual layer rings, and a two-part polyurethane top-coat (50 μm DFT, UV-stable, RAL 9003) provided colour and solar protection.

D. Internal outfitting

The internal layout relies on a laser-cut 2 mm 6061-T6 aluminum baseplate that slides on two printed guide rails and locks with four M6 captive bolts. The ESC stack, STM32 control board, and Jetson Nano sit on rubber isolators to damp vibration. A 3 mm aluminum keel strip bonded beneath the hull forms the main structural spine. Fore and aft ballast pockets each hold 25 g of sintered steel pellets – yielding –14 g buoyancy when loaded and neutral buoyancy after removing two pellets. All sensor and camera ports use 5 mm polycarbonate windows compressed on dash-118 EPDM O-rings per Parker Table 4-2.

E. Functional shakedown

A static vacuum soak at –80 kPa for 12 h showed zero pressure rise. Two cold-tank immersions to 2.5 bar ($\approx 25 \text{ m}$) likewise revealed no leaks or crazing. Deck tests verified stable trim within $\pm 1.1^\circ$ pitch and $\pm 0.9^\circ$ roll.

Four reef-flank sorties off Kalpitiya – covering 37 km over three days – confirmed operational readiness: solvent joints

held, the ABS surface resisted coral brush, and on-board temperatures remained below 45 °C in 31 °C ambient water. OpenLCA modelling, using Ecoinvent 3.9 datasets, showed the ABS build carried 18 % less embodied energy and 65 % lower VOC emissions than an equivalently sized hand-lay-up fiberglass hull produced for control testing.

F. Observed challenges and mitigation

Achieving a blemish-free surface required two cataloy passes despite vapour smoothing; printing with concentric skins instead of gyroid on exterior layers reduced visible texture by $\sim 25 \%$.

Solvent-bond cure time was humidity-sensitive; pre-heating the joint area proved easier than tenting the full shell.

Fine thread engagement in ABS inserts loosened after 10-h vibration tests; switching to brass heat-sets solved the issue.

Figure 4 collates the build sequence—from raw prints through bonded shell, painted hull, and the completed vehicle poised for mission deployment—demonstrating that an exclusively ABS, large-format-printed construction can meet the structural, logistical, and environmental requirements of shallow-reef drones without recourse to fiberglass or metal tooling.

VII. CONCLUSION

This paper details the additive manufacturing process for an ABS-based shallow-sea drone hull, emphasizing the design and integration of a structurally sound internal baseplate. Results from mechanical loading and pressurized submersion trials indicate that with proper layer orientation, bonding techniques, and epoxy sealing, 3D-printed ABS structures can withstand moderate water pressures and remain leak-free. The success of the baseplate ensures stable support for essential electronics, while the specialized camera mount enables secure, user-friendly imaging tasks. Ultimately, ABS 3D printing represents a viable, economical alternative for near-shore drone construction, accelerating development cycles while maintaining reliability in the field.

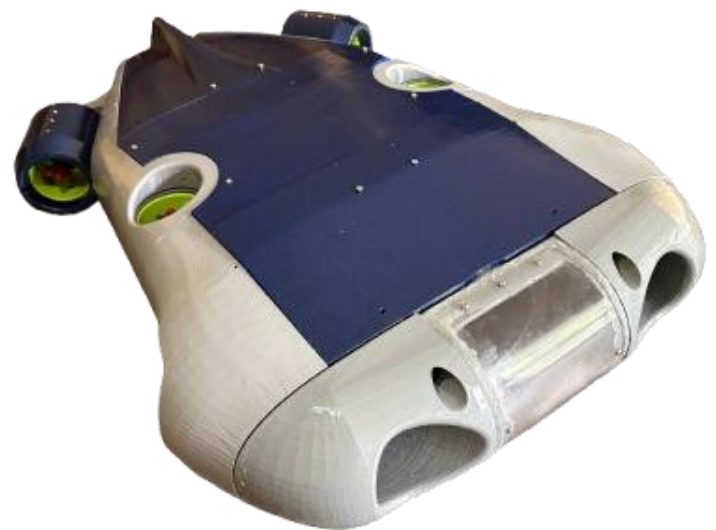


Fig. 4. Final Fabricated Underwater drone.

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